12_evaluation_depth_comparison

August 31, 2026

[1]:
```r
library(Seurat)

library(reticulate)
library(anndata)

library(ggplot2)
library(ggrepel)
library(ggpubr)
library(ggpattern)
library(pheatmap)
library(dplyr)
library(tidyr)
library(RColorBrewer)
library(clustree)
library(repr)
library(patchwork)
options(repr.plot.width=10, repr.plot.height=8)


library(UpSetR)
library(grid)

library(PRROC)
library(Matrix)

getwd()

dataset_id <- "simulated_mm_RA"
genome_id <- "mm10"
samples <- c("old", "young")
colorSamples <- c("old"="#7a646a",
                  "young"="#88bce7")

dir.create(paste0("figures_", dataset_id))
for (sample in samples) {
    dir.create(paste0("figures_", dataset_id, "_", sample))
}
```

```r
colorTools <- c( # "MATES"="#F98A7B",
    "STARsolo_TE" = "#FFBC81",
    "STARsolo_TE_EM" = "#F3E088",
    "SoloTE_unique" = "#A4DD9B",
    # "SoloTE_thr2" = "#6FC69D",
    # "SoloTE_thr1" = "#4BB2BB",
    # "SoloTE_thr0" = "#3989BF",
    "Stellarscope" = "#4F5D93",
    "simulated" = "grey70"
)

thrMinCells <- 500 * 0.05
```

```
Loading required package: SeuratObject

Loading required package: sp

'SeuratObject' was built under R 4.3.3 but the current version is
4.4.1; it is recomended that you reinstall 'SeuratObject' as the ABI
for R may have changed

'SeuratObject' was built with package 'Matrix' 1.6.4 but the current
version is 1.7.5; it is recomended that you reinstall 'SeuratObject' as
the ABI for 'Matrix' may have changed


Attaching package: 'SeuratObject'


The following objects are masked from 'package:base':

    intersect, t



Attaching package: 'anndata'


The following object is masked from 'package:SeuratObject':

    Layers


Warning message:
"package 'ggpattern' was built under R version 4.4.3"
Warning message:
```

```
"package ‘dplyr’ was built under R version 4.4.3"

Attaching package: ‘dplyr’


The following objects are masked from ‘package:stats’:

    filter, lag


The following objects are masked from ‘package:base’:

    intersect, setdiff, setequal, union


Warning message:
"package ‘tidyr’ was built under R version 4.4.3"
Warning message:
"package ‘RColorBrewer’ was built under R version 4.4.3"
Loading required package: ggraph


Attaching package: ‘ggraph’


The following object is masked from ‘package:sp’:

    geometry


Warning message:
"package ‘patchwork’ was built under R version 4.4.3"
Loading required package: rlang

Warning message:
"package ‘rlang’ was built under R version 4.4.3"

Attaching package: ‘Matrix’


The following objects are masked from ‘package:tidyr’:

    expand, pack, unpack
```

'/mnt/TEdata/TEbenchmarking/Rscripts'

```
Warning message in dir.create(paste0("figures_", dataset_id)):
```

```
"'figures_simulated_mm_RA' already exists"
Warning message in dir.create(paste0("figures_", dataset_id, "_", sample)):
"'figures_simulated_mm_RA_old' already exists"
Warning message in dir.create(paste0("figures_", dataset_id, "_", sample)):
"'figures_simulated_mm_RA_young' already exists"
```

[2]:
```
gc()
```

A matrix: 2 × 6 of type dbl

	used	(Mb)	gc trigger	(Mb)	max used	(Mb)
Ncells	3618362	193.3	6753332	360.7	5192211	277.3
Vcells	6278504	48.0	12255594	93.6	9459355	72.2

1 Load seurat objects

[3]:
```
load("workspaces/00_evaluation_objectCreation.Rdata")

colorTools <- c( # "MATES"="#F98A7B",
    "STARsolo_TE" = "#FFBC81",
    "STARsolo_TE_EM" = "#F3E088",
    "SoloTE_unique" = "#A4DD9B",
    "SoloTE_thr2" = "#6FC69D",
    "SoloTE_thr1" = "#4BB2BB",
    "SoloTE_thr0" = "#3989BF",
    "Stellarscope" = "#4F5D93",
    "Stellarscope_manualrun" = "#2E3D72",
    "simulated" = "grey70"
)
```

[4]:
```
dataset_ids <- c("simulated_mm_RA_5kRpC",
    "simulated_mm_RA_20kRpC",
    "simulated_mm_RA_50kRpC")
depths <- c("5kRpC", "20kRpC", "50kRpC", "100kRpC")
names(depths) <- dataset_ids

objList_byDepth <- list()

for(id in dataset_ids){

    objList_byDepth[[as.character(depths[id])]] <- readRDS(paste0("data/
 ↪objList_", id, ".RDS"))
}

objList_byDepth[["100kRpC"]] <- objList[names(objList_byDepth[["50kRpC"]])]

names(objList_byDepth)
names(objList_byDepth[["100kRpC"]])
```

1. '5kRpC' 2. '20kRpC' 3. '50kRpC' 4. '100kRpC'

1. 'STARsolo_TE' 2. 'STARsolo_TE_EM' 3. 'SoloTE_unique' 4. 'Stellarscope'

2 Correlations

2.0.1 TP COUNTS

[4]:
```
library(data.table)

layer <- "data"
results_list <- list()

for (sample in samples) {

    splatter_obj <- splatter_objs[[sample]]
    splatterMat_full <- GetAssayData(splatter_obj, layer = layer)   # computed␣
↪once per sample
    splatter_features <- Features(splatter_obj)

    for (depth in depths) {

        for (tool in names(objList_byDepth[[depth]])) {

            obj <- objList_byDepth[[depth]][[tool]][[sample]]

            TP_TEs <- intersect(splatter_features, Features(obj))

            splatter_subset <- as.matrix(splatterMat_full[TP_TEs, , drop =␣
↪FALSE])
            tool_subset     <- as.matrix(GetAssayData(obj, layer =␣
↪layer)[TP_TEs, , drop = FALSE])

            # vectorized melt
            dt <- data.table(
                TE         = rep(TP_TEs, times = ncol(splatter_subset)),
                Cell       = rep(colnames(splatter_subset), each =␣
↪nrow(splatter_subset)),
                sim_count  = as.vector(splatter_subset),
                tool_count = as.vector(tool_subset)
            )

            dt <- dt[sim_count > 0 & tool_count > 0]

            dt[, order := conversion_table$class[match(TE,␣
↪conversion_table$stellarscopeID)]]
            dt[, `:=`(tool = tool, sample = sample, depth = depth)]
```

```
            results_list[[paste(sample, depth, tool, sep = "_")]] <- dt
        }
    }
}

densityPlotsDf <- rbindlist(results_list)
```

```
Warning message:
"package ‘data.table’ was built under R version 4.4.3"

Attaching package: ‘data.table’


The following object is masked from ‘package:rlang’:

    :=


The following objects are masked from ‘package:dplyr’:

    between, first, last
```

[5]:
```
densityPlotsDf$family <- conversion_table$family[match(densityPlotsDf$TE,
 ↪conversion_table$stellarscopeID)]
densityPlotsDf$subfamily <- conversion_table$subfamily[match(densityPlotsDf$TE,
 ↪conversion_table$stellarscopeID)]
densityPlotsDf$propSubs <- conversion_table$propSubs[match(densityPlotsDf$TE,
 ↪conversion_table$stellarscopeID)]
```

[6]:
```
# group by TE, depth and tool and compute spearman correlations, only for loci
 ↪detected in at least 25 cells

cor_by_TE <- densityPlotsDf %>%
  group_by(TE, tool, sample, depth) %>%
  filter(n() >= thrMinCells) %>%
  summarise(
    order  = first(order),
    family = first(family),
    subfamily = first(subfamily),
    n = n(),
    cor = cor(sim_count, tool_count, method="spearman", use = "complete.obs"),
    p_value = cor.test(sim_count, tool_count, method = "spearman", use =
 ↪"complete.obs")$p.value,
    .groups = "drop"
```

```
)
```

```
Warning message:
"There were 40122 warnings in `summarise()`.
The first warning was:
  In argument: `p_value = cor.test(sim_count, tool_count,
method = "spearman",
   use = "complete.obs")$p.value`.
  In group 6: `TE = "AmnSINE1-dup1041"`, `tool = "STARsolo_TE"`,
`sample =
   "old"`, `depth = "20kRpC"`.
Caused by warning in `cor.test.default()`:
! Cannot compute exact p-value with ties
  Run `dplyr::last_dplyr_warnings()` to see the 40121
remaining warnings."
```

[7]:
```
head(cor_by_TE)
```

A tibble: 6 × 10

TE <chr>	tool <chr>	sample <chr>	depth <chr>	order <chr>	family <chr>	subfamily <chr>
AmnSINE1-dup1015	STARsolo_TE	old	100kRpC	SINE	Deu	AmnSINE
AmnSINE1-dup1015	STARsolo_TE_EM	old	100kRpC	SINE	Deu	AmnSINE
AmnSINE1-dup1015	SoloTE_unique	old	100kRpC	SINE	Deu	AmnSINE
AmnSINE1-dup1015	Stellarscope	old	100kRpC	SINE	Deu	AmnSINE
AmnSINE1-dup1041	STARsolo_TE	old	100kRpC	SINE	Deu	AmnSINE
AmnSINE1-dup1041	STARsolo_TE	old	20kRpC	SINE	Deu	AmnSINE

[8]:
```
colorTools <- c( # "MATES"="#F98A7B",
    "STARsolo_TE" = "#FFBC81",
    "STARsolo_TE_EM" = "#F3E088",
    "SoloTE_unique" = "#A4DD9B",
    # "SoloTE_thr2" = "#6FC69D",
    # "SoloTE_thr1" = "#4BB2BB",
    # "SoloTE_thr0" = "#3989BF",
    "Stellarscope" = "#4F5D93",
    "simulated" = "grey70"
)
```

[9]:
```
cor_by_TE <- cor_by_TE[cor_by_TE$tool %in% names(colorTools),]

# order tools to plot
cor_by_TE$tool <- factor(cor_by_TE$tool , levels = rev(names(colorTools)))

cor_by_TE$depth <- factor(cor_by_TE$depth, levels = c("5kRpC", "20kRpC",␣
↪"50kRpC", "100kRpC"))

# add other covariates
```

```r
cor_by_TE$mya <- conversion_table$mya[match(cor_by_TE$TE,␣
↪conversion_table$stellarscopeID)]
cor_by_TE$propSubs <- conversion_table$propSubs[match(cor_by_TE$TE,␣
↪conversion_table$stellarscopeID)]
cor_by_TE$TElength <- conversion_table$end[match(cor_by_TE$TE,␣
↪conversion_table$stellarscopeID)] -
                                                    ␣
↪conversion_table$start[match(cor_by_TE$TE, conversion_table$stellarscopeID)]

head(cor_by_TE)
```

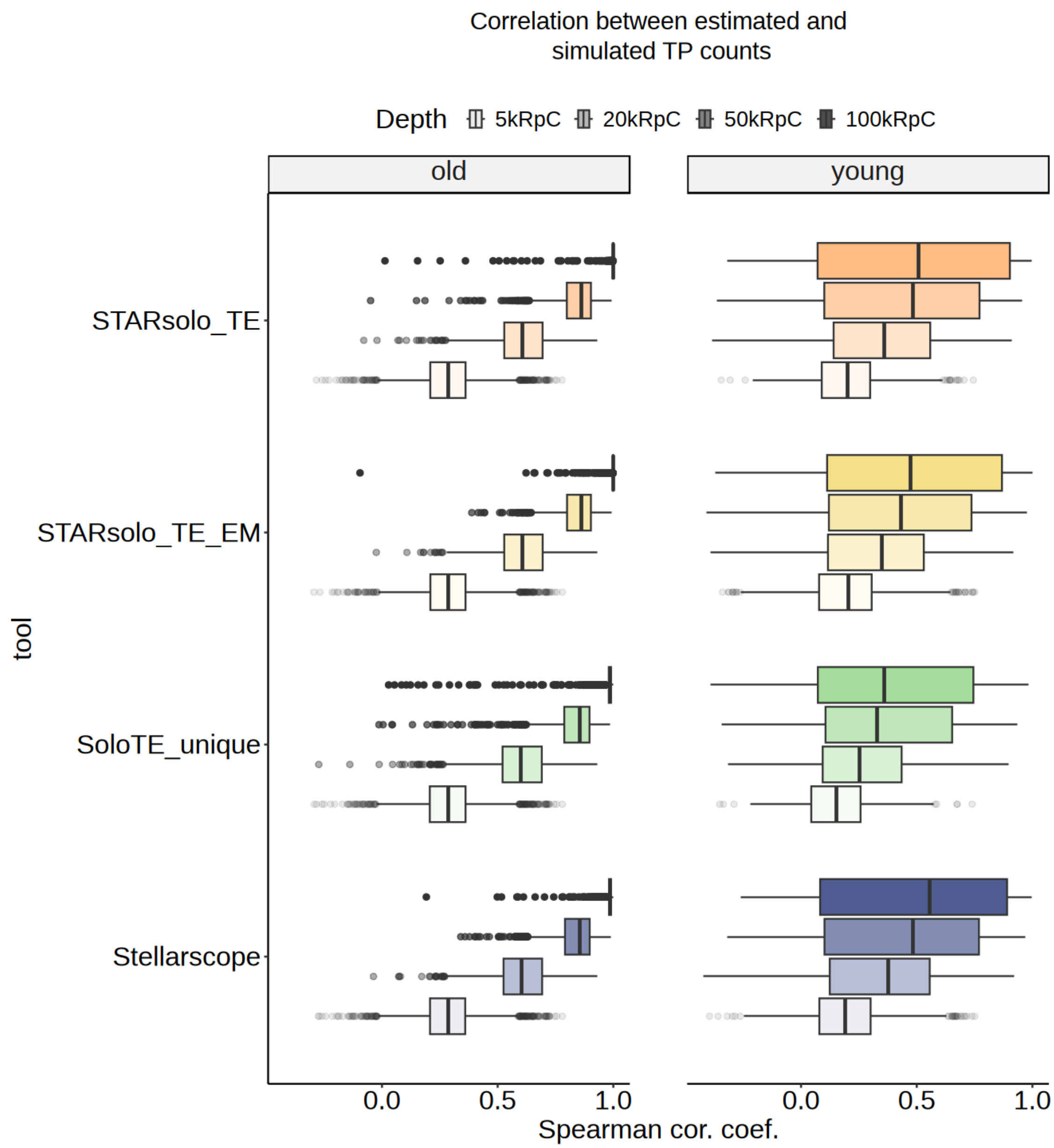
A tibble: 6 × 13

TE <chr>	tool <fct>	sample <chr>	depth <fct>	order <chr>	family <chr>	subfamily <chr>
AmnSINE1-dup1015	STARsolo_TE	old	100kRpC	SINE	Deu	AmnSINE
AmnSINE1-dup1015	STARsolo_TE_EM	old	100kRpC	SINE	Deu	AmnSINE
AmnSINE1-dup1015	SoloTE_unique	old	100kRpC	SINE	Deu	AmnSINE
AmnSINE1-dup1015	Stellarscope	old	100kRpC	SINE	Deu	AmnSINE
AmnSINE1-dup1041	STARsolo_TE	old	100kRpC	SINE	Deu	AmnSINE
AmnSINE1-dup1041	STARsolo_TE	old	20kRpC	SINE	Deu	AmnSINE

[22]:
```r
options(repr.plot.width=11, repr.plot.height=12)

ggplot(cor_by_TE[cor_by_TE$sample %in% c("old","young"),]) +
        aes(x=cor, fill=tool,  y=tool, alpha=depth) +
  facet_wrap(~sample, ncol=2) +
  geom_boxplot( linewidth = 0.5, ) +
  ggtitle("Correlation between estimated and",
          subtitle = " simulated TP counts\n") +
  xlab("Spearman cor. coef.") +
  scale_fill_manual(values = colorTools) +
  theme_pubr() +
  guides(fill="none", color="none", alpha=guide_legend(title="Depth", override.
↪aes=list(fill="grey30"))) +
  theme(text=element_text(size=20), axis.text.y = element_text(size=20),
        plot.title = element_text(size=18, hjust=0.5),
        plot.subtitle = element_text(size=18, hjust=0.5),
        strip.text = element_text(size=20, hjust=0.5, vjust = 1 ),
        panel.spacing = unit(3, "lines"))
ggsave(paste0("figures_",dataset_id,"/
↪correlation_perTE_boxplot_byAge_byTool_byDepth_TPcounts.pdf"),width=11,␣
↪height=12)
```

```
Warning message:
"Using alpha for a discrete variable is not advised."
Warning message:
"Using alpha for a discrete variable is not advised."
```

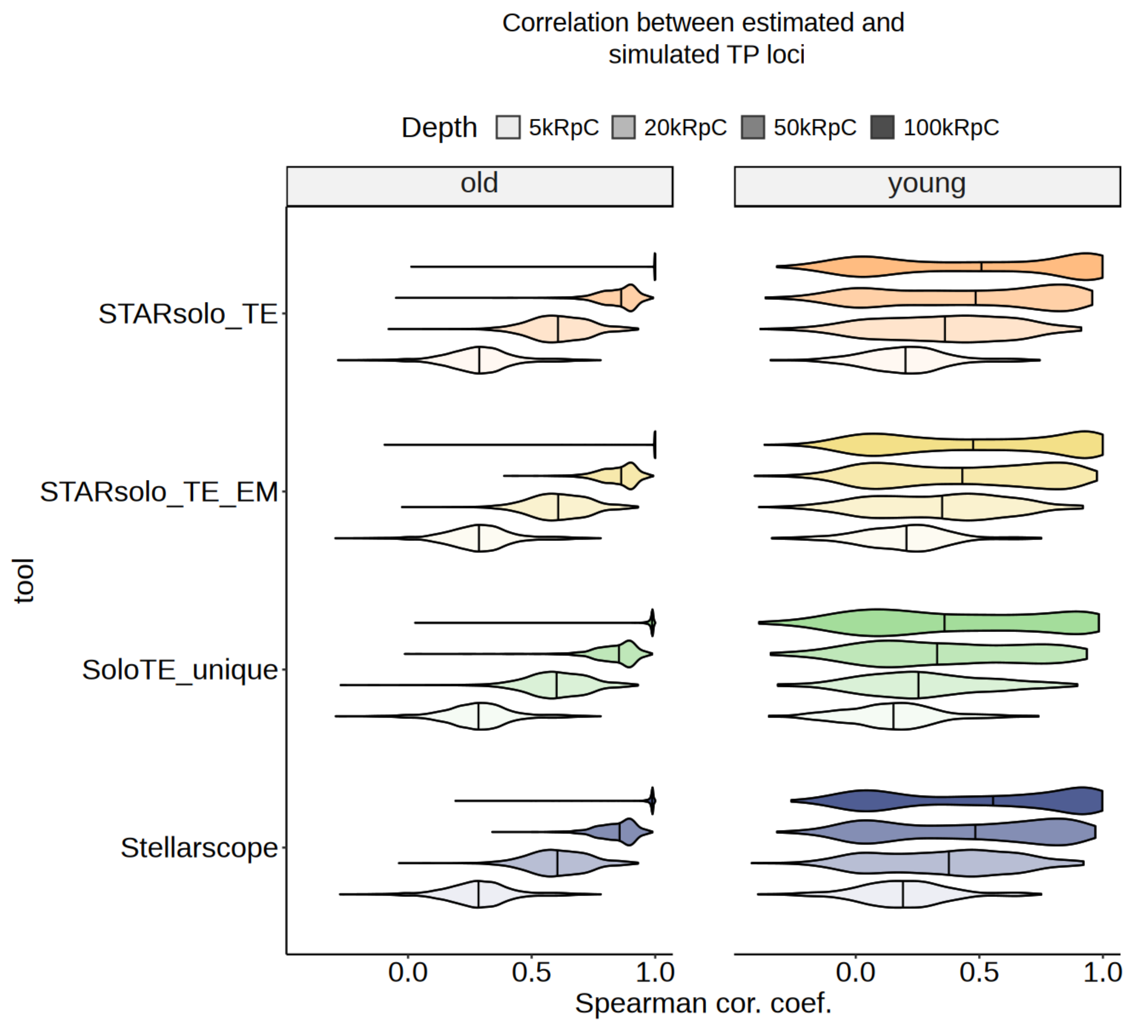


[21]:
```
options(repr.plot.width=11, repr.plot.height=10)

ggplot(cor_by_TE[cor_by_TE$sample %in% c("old","young"),]) +
        aes(x=tool, y=cor, fill=tool, alpha=depth) +
  facet_wrap(~sample, ncol=2) +
  geom_violin(linewidth = 0.5, scale = "width", width = 0.6,
              position = position_dodge(width = 0.7)) +
  geom_violin(aes(group = interaction(tool, depth)),
              fill = NA, alpha = 1, color = "black",
              linewidth = 0.5, scale = "width", width = 0.6,
```

```
              draw_quantiles = c(0.5),
              position = position_dodge(width = 0.7),
              show.legend = FALSE) +
  coord_flip() +
  scale_alpha_discrete() +
  ggtitle("Correlation between estimated and",
          subtitle = " simulated TP loci\n") +
  ylab("Spearman cor. coef.") +
  xlab("tool") +
  scale_fill_manual(values = colorTools) +
  theme_pubr() +
  guides(fill="none", color="none", alpha=guide_legend(title="Depth", override.
 ↪aes=list(fill="grey30"))) +
  theme(text=element_text(size=20), axis.text.y = element_text(size=20),
        plot.title = element_text(size=18, hjust=0.5),
        plot.subtitle = element_text(size=18, hjust=0.5),
        strip.text = element_text(size=20, hjust=0.5, vjust = 1 ),
        panel.spacing = unit(3, "lines"))
ggsave(paste0("figures_",dataset_id,"/
 ↪correlation_perTE_violin_byAge_byTool_byDepth_TPcounts.pdf"),width=11,␣
 ↪height=10)
```

```
Warning message:
"Using alpha for a discrete variable is not advised."
```



Correlation between estimated and
simulated TP loci
Depth
5kRpC
20kRpC
50kRpC
100kRpC
old
young
STARsolo_TE
STARsolo_TE_EM
SoloTE_unique
Stellarscope
tool
0.0
0.5
1.0
0.0
0.5
1.0
Spearman cor. coef.